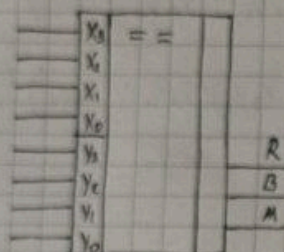


Контрольная работа Компаратор

2. УГО



1. Табл. истинности

X ₃ X ₂ X ₁ X ₀	Y ₃ Y ₂ Y ₁ Y ₀	R B M
0 0 0 0	0 0 0 0	1 0 0
1 * * *	0 * * *	0 1 0
0 * * *	1 * * *	0 0 1
0 1 * *	0 0 * *	0 1 0
0 0 * *	0 1 * *	0 0 1
1 1 * *	1 0 * *	0 1 0
1 0 * *	1 1 * *	0 0 1
0 0 1 *	0 0 0 *	0 1 0
0 0 0 *	0 0 1 *	0 0 1
1 1 1 *	1 1 0 *	0 1 0
1 1 0 *	1 1 1 *	0 0 1
1 0 1 *	1 0 0 *	0 1 0
1 0 0 *	1 0 1 *	0 0 1
0 1 1 *	0 1 0 *	0 1 0
0 1 0 *	0 1 1 *	0 0 1
0 0 0 1	0 0 0 0	0 1 0
0 0 0 0	0 0 0 1	0 0 1
1 1 1 1	1 1 1 0	0 1 0
1 1 1 0	1 1 1 1	0 0 1
0 0 1 1	0 0 1 0	0 1 0
0 0 1 0	0 0 1 1	0 0 1
0 1 0 1	0 1 0 0	0 1 0
0 1 0 0	0 1 0 1	0 0 1
1 0 0 1	1 0 0 0	0 1 0
1 0 0 0	1 0 0 1	0 0 1
0 1 1 1	0 1 1 0	0 1 0
0 1 1 0	0 1 1 1	0 0 1
1 0 1 1	1 0 1 0	0 1 0
1 0 1 0	1 0 1 1	0 0 1
1 1 0 1	1 1 0 0	0 1 0
1 1 0 0	1 1 0 1	0 0 1

2. КУ

$$R = (\bar{x}_3 \bar{y}_3 \vee x_3 y_3) (\bar{x}_2 \bar{y}_2 \vee x_2 y_2) (\bar{x}_1 \bar{y}_1 \vee x_1 y_1) (\bar{x}_0 \bar{y}_0 \vee x_0 y_0)$$

$$B = (\bar{x}_3 \bar{y}_3) \vee (\bar{x}_3 \bar{y}_3 \vee x_3 y_3) (\bar{x}_2 \bar{y}_2) \vee (\bar{x}_3 \bar{y}_3 \vee x_3 y_3) (\bar{x}_2 \bar{y}_2 \vee x_2 y_2) (\bar{x}_1 \bar{y}_1) \vee (\bar{x}_3 \bar{y}_3 \vee x_3 y_3) (\bar{x}_2 \bar{y}_2 \vee x_2 y_2) (\bar{x}_1 \bar{y}_1) (\bar{x}_0 \bar{y}_0)$$

$$M = (\bar{x}_3 \bar{y}_3) \vee (\bar{x}_3 \bar{y}_3 \vee x_3 y_3) (\bar{x}_2 \bar{y}_2) \vee (\bar{x}_3 \bar{y}_3 \vee x_3 y_3) (\bar{x}_2 \bar{y}_2 \vee x_2 y_2) (\bar{x}_1 \bar{y}_1) \vee (\bar{x}_3 \bar{y}_3 \vee x_3 y_3) (\bar{x}_2 \bar{y}_2 \vee x_2 y_2) (\bar{x}_1 \bar{y}_1) (\bar{x}_0 \bar{y}_0)$$

3. Минимизация —

R — не имеет повторов

B — невозможно минимизировать, так как они состоят с парами

M — аналогично B

и функциональная схема (на другой стр.)

Ullrich

